



Quantum convolutional neural network based on variational quantum circuits

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ABSTRACT

Machine learning algorithms are becoming increasingly resource-intensive. In contrast to classical computing, quantum computing holds the potential with exponential advantages and promises to overcome the bottleneck in computing performance. Taking inspiration from the convolution and pooling operations of convolutional neural networks, a quantum convolutional neural network (QCNN) is designed based on pure variational quantum circuits. By combining the advantages of angle encoding and amplitude encoding, a tree-structured hybrid amplitude encoding method has been developed. This method provides flexibility and stability in adjusting the width and the depth of the quantum circuit required for data encoding. Experiments conducted on the MNIST dataset and the Fashion-MNIST dataset demonstrate that the QCNN achieves faster convergence speed and higher image classification accuracy than the classical convolutional neural networks. Specifically, the QCNN reaches convergence after just 50 iterations, whereas the classical counterpart requires more than 200 iterations to converge. Furthermore, when performing the PCA dimensionality reduction to 8, the QCNN with Circuit 6 achieves an accuracy of 98.54% on the MNIST dataset, while the classical counterpart achieves only 86.32%.

1. Introduction

Image classification has always been a critical application field of supervised learning, and finds extensive use in the fields such as medical diagnosis [1] and face recognition [2]. The image classification process typically consists of two main stages: feature extraction and classification based on the extracted features. Before the emergence of convolutional neural networks, feature extraction was primarily achieved through techniques like scale-invariant feature transformation [3], SURF algorithm [4], and FAST algorithm [5]. Subsequently, these extracted features are used as inputs to classifiers such as K-nearest neighbor models [6], support vector machines, and decision trees [7].

CNNs, like other machine learning algorithms, face the problem of ever-expanding data scale. Therefore, many scholars have embarked on the integration of quantum computing with CNNs, using quantum computing to improve the performance of CNNs. In 2019, extending the main features of CNNs to quantum systems, Cong et al. designed a QCNN to avoid the exponential data increase when to transfer quantum data into classical computing environment [8]. In 2020, Mari et al. introduced the classical transfer learning method into the quantum domain, using the classical convolutional layers trained in the classical deep

neural network for image feature extraction, and then training the quantum neural network separately to learn the decision boundary from these features [9]. In the same year, Oh et al. trained a hybrid quantum-classical CNN with prediction accuracy close to that of the classical CNNs [10]. Subsequently, Oh et al. investigated QCNNs based on quantum random access memory [11]. In 2021, Liu et al. designed a QCNN for classical image classification with a hybrid quantum-classical structure, and verified its classification performance with the Tetris dataset [12]. In 2021, Pesah et al. conducted a rigorous analysis of gradient scaling for parameters in QCNN, demonstrated a key problem that is easy to encounter in quantum neural network training, i.e., barren plateau, will not appear in QCNN, and conducted numerical experiments to verify its conclusion [13]. In 2022, MacCormack et al. introduced a novel branching QCNN, which enables to train a higher number of parameters within shallow quantum circuits, primarily focusing on the classification of the simplest quantum states [14]. In 2023, Gao et al. proposed a quantum algorithm for kernelized correlation filter based on the special properties of circulant matrices [15].

Since the convolutional kernels can extract the local features of image data, CNN has become one of the preferred machine learning algorithms in the field of image classification. Convolutional layers

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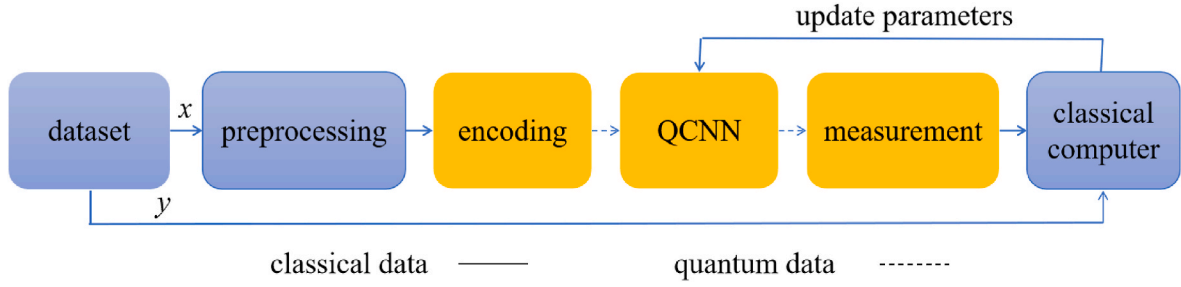


Fig. 1. Basic process of QCNN.

based on variational quantum circuits can establish global entanglement, which can not only capture local features of data, but also have the potential to extract global features. Therefore, compared with CNNs, QCNNs based on variational quantum circuits not only offer the potential for quantum acceleration but also excel in global feature extraction. Consequently, studying QCNNs for image classification holds substantial significance.

However, the depth and the width are two sides of a quantum circuit. A deeper quantum circuit means more qubits, which means the feasibility of classical simulation; A wider quantum circuit means that there are enough qubits, and then better entanglement ability will be obtained. Therefore, it is necessary to develop a coding method capable of adjusting the depth and the width of the quantum circuit to optimize the training effectiveness of QCNNs.

The main contributions of this work are:

- (a) A hybrid tree structured amplitude encoding was studied by combining amplitude encoding with tree shaped encoding.
- (b) In image recognition, the data is dimensionally reduced by PCA and Autoencoder.
- (c) QCNN models with convolutional layer and pooling layer are designed and trained under the MNIST and the Fashion MNIST datasets.

The rest of this paper is organized as follows. Section 2 introduces some fundamental knowledge on variational quantum circuit. Section 3 gives the hybrid tree structured amplitude encoding and the quantum convolutional neural network based on pure variational quantum circuits. Section 4 presents the experiments on the MNIST and Fashion-MNIST datasets. Section 5 concludes this paper.

2. Variational quantum circuit

In 2007, Sousa et al. proposed a general quantum circuit to achieve any specific function by setting appropriate parameters [16]. This kind of universal programmable quantum circuit is called variational quantum circuit or parametric quantum circuit. Variational quantum circuit generally consists of the following three parts:

- (1) A fixed initial state, usually in the state $|1\rangle$;
- (2) Quantum circuit $U(\theta)$ composed of a series of quantum gates, some or all quantum gates contain free parameters, θ is a set of free parameters;
- (3) The measurement operation $\hat{B}$ used to obtain the observable measurement can be performed at the end of one or more quantum circuits.

The expectation value of the variational quantum circuit, measured after the quantum state collapses, can be represented as [17]

$$f(\theta) = \langle 0 | U^\dagger(\theta) \hat{B} U(\theta) | 0 \rangle \quad (1)$$

When using a variational quantum circuit for processing tasks, both

the free parameter set θ and a set of fixed parameters $x = (x_1, x_2, \dots)$ can be fed into the quantum circuit as the parameters of quantum gates. This allows classical information x to be transformed into quantum information for processing. The process of embedding classical information x into a variational quantum circuit is referred to as data encoding. After incorporating the fixed parameter set x , the expectation value of the variational quantum circuit can be represented as

$$f(x; \theta) = \langle 0 | U^\dagger(x; \theta) \hat{B} U(x; \theta) | 0 \rangle \quad (2)$$

Classical data is input into the quantum circuit through data encoding. Subsequently, by conducting measurement operations on the quantum circuit, classical information can be extracted. This approach represents one of the methods to process classical data using variational quantum circuits.

3. Quantum convolutional neural network

3.1. Basic algorithm process

Currently, most quantum machine learning algorithms undergo optimization through collaboration between quantum computers and classical ones. The main body of the QCNN structure proposed here is composed of pure variable component sub circuits, and the parameter optimization part is executed in cooperation with a classical computer [18].

Therefore, the QCNN algorithm proposed entails a preliminary dimension reduction of classical image data before encoding it into a quantum circuit. It is worth noting that the algorithm is primarily validated for binary classification task. The basic flow of the QCNN is shown in Fig. 1, where y represents the corresponding label for data x . solid lines represent classical data, while dotted lines depict quantum data. The yellow frames handle quantum data, while the blue frames process classical data.

The classical image data undergoes preprocessing prior to its encoding into the quantum circuit, and the preprocessed data is transformed into quantum data with the data encoding circuit. The quantum convolution layer composed of a set of quantum gates extracts the features of quantum data, and then the dimension of the feature data is reduced by the pooling layer. One of the key characteristics of CNNs, translation invariance, is also preserved in QCNN. Therefore, the parameters used for convolution operations on different qubits in the same convolutional layer are same too. Towards the end of the QCNN model, information is extracted into a single qubit and the predicted results are obtained by measuring that qubit. The gradient of the quantum circuit is calculated with the parameter shift rule, and the classical computer updates the parameters of the QCNN model based on the quantum gradient, so as to improve the prediction accuracy further.

3.2. Data encoding in the variational quantum circuit

Quantum data encoding is an essential step in converting classical data into a quantum state. Therefore, selecting an appropriate encoding

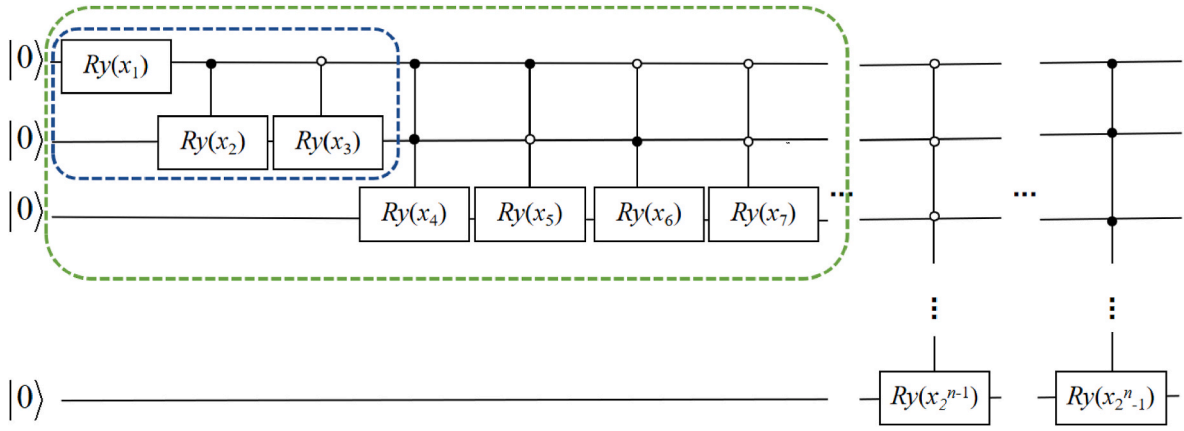


Fig. 2. Tree-structured hybrid amplitude encoding circuit.

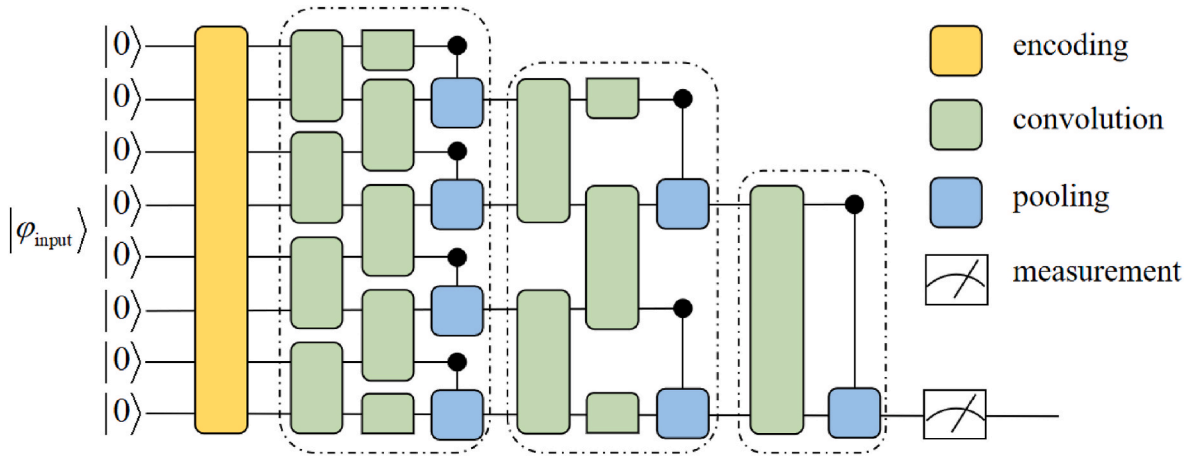


Fig. 3. Schematic of the QCNN circuit.

method is of paramount importance to fully harness the acceleration potential of quantum machine learning algorithms. It was reported that quantum data encoding has at least exponential complexity in the worst case [19]. LaRose et al. introduced several commonly used encoding methods, including angle encoding, dense angle encoding, and amplitude encoding [20].

Angle coding and dense angle coding are advantageous for reducing the depth of quantum circuit. However, these two encoding methods require $O(M)$ order of magnitude qubits to encode the classical data with dimension M , whereas the NISQ devices can only provide a limited number of qubits. In contrast, amplitude encoding is beneficial for reducing the qubit requirement. To encode classical data with dimension M with amplitude encoding, it only requires qubits of order of $O(\log M)$ [21]. However, the depth of the quantum circuit has reached the order of $O(M)$. In the NISQ devices, a too deep circuit will destruct the quantum state by noise and decoherence. These two methods represent somewhat extreme options. To strike a balance between circuit depth and qubit number, a combination of these two encoding methods was adopted.

N qubits were divided into m independent blocks of n -qubit each, and each block is independently encoded by amplitude encoding. Each block can encode 2^n qubits of classical data, and the entire N qubit circuit can encode $m2^n$ qubits of classical data. This hybrid amplitude coding method allows to adjust the values of m and n based on the dimension M of classical data and the limit of quantum equipment in the actual task, so as to adjust the width and the depth of the variable component sub circuit. When n takes its minimal value of 1, the depth of quantum circuit is $2^1 = 2$, and the width is $M/2$ qubits; Conversely, when n assumes its

maximal value of N , hybrid amplitude encoding is equivalent to amplitude encoding.

However, it's important to note that while the hybrid amplitude encoding method offers great flexibility, it can also introduce inaccuracies during the encoding process. This is because each qubit region independently normalizes its encoded classical data, potentially leading to inconsistent scaling ratios among different blocks for the classical data. Consequently, the encoded data may not accurately represent the original data.

Inspired by the concept of balanced controlled rotation operations introduced by Mottonen et al. [22], a novel data encoding method called tree-structured hybrid amplitude encoding is proposed. This method aims to strike a balance between the depth of the quantum circuit and the number of qubits while ensuring the accurate representation of information. The idea of tree-structured hybrid amplitude encoding shares similarities with the hybrid amplitude encoding, which divides N qubits into m independent blocks of n qubits each, with each region being encoded independently. However, the encoding method within each block is different from that of the ordinary amplitude encoding. The tree amplitude encoding process in an n -qubit area is shown in Fig. 2, which is primarily executed through rotation gates or controlled rotation gates. The tree amplitude encoding for two qubits is enclosed in the blue dashed frame, while the tree amplitude encoding for three qubits is enclosed in the green dashed frame. There is $2^{1-1} = 1$ rotation gate with parameters acting on the first qubit circuit, $2^{2-1} = 2$ rotation gates acting on the second qubit circuit, and there are 2^{n-1} rotation gates acting on the n -th qubit circuit. These rotation gates operate the

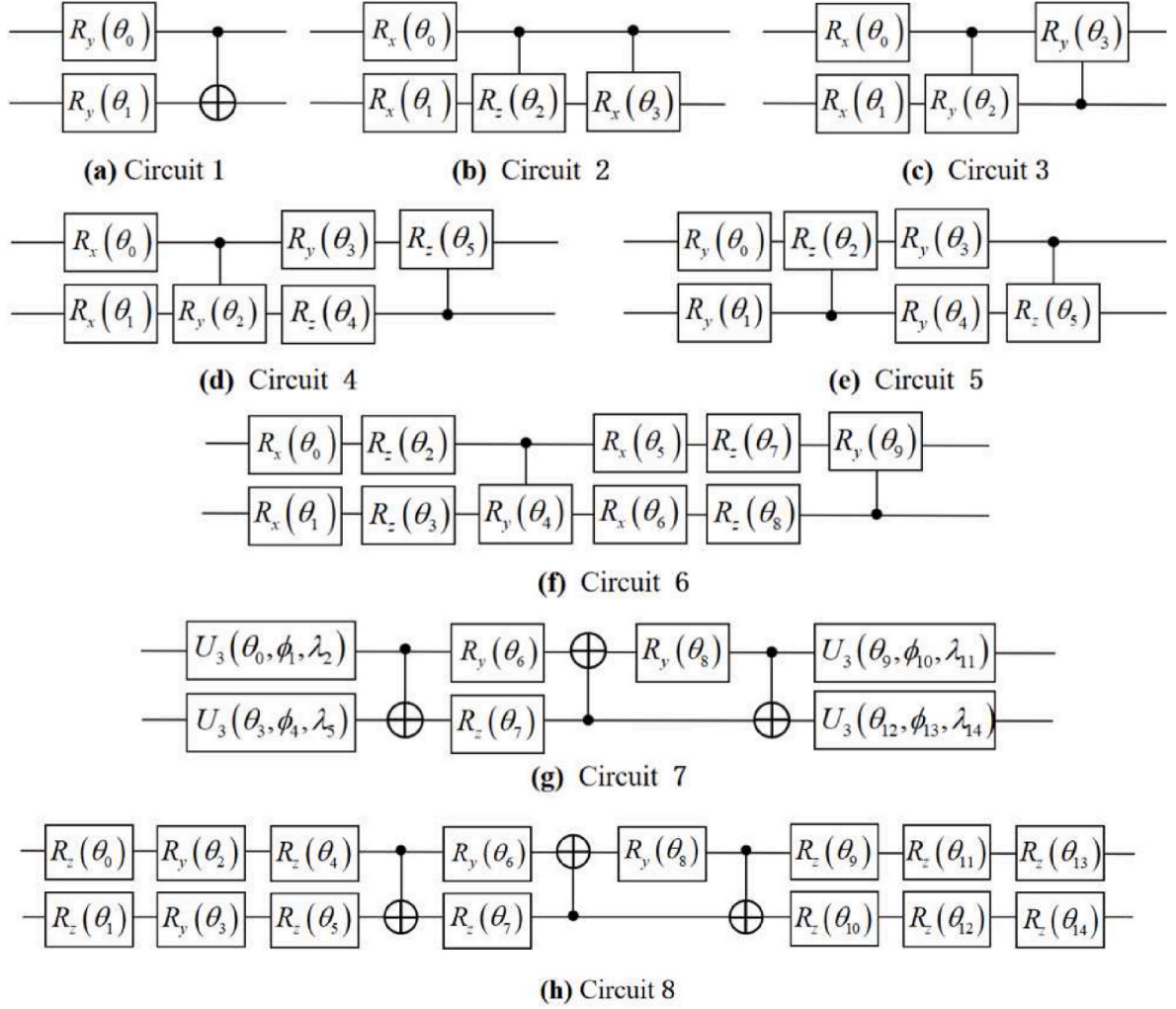


Fig. 4. Circuit for quantum convolutional layer.

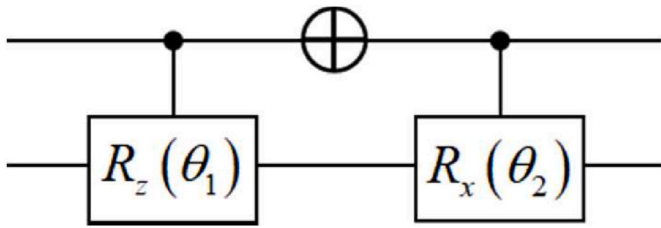


Fig. 5. Circuit for pooling layer.

amplitude through different control conditions. The control conditions of each rotation gate in the n -th qubit circuit are $|s_1\rangle, \dots, |s_l\rangle, \dots, |s_{2^n-1}\rangle$, where s_l is the $n-1$ bit binary representation corresponding to integer l . The entire encoding process can be represented as

$$\left(\prod_{i=1}^n \prod_{j=1}^{2^{i-1}} U_{ij}(x_{ij}) \right) |0\rangle^{\otimes n} \quad (3)$$

where the subscript i represents the i -th qubit circuit, and the subscript j represents the j -th unitary operation of the circuit, U_{ij} is the corresponding unitary operation, which can be a single quantum gate or a combination of multiple quantum gates. In variational quantum circuits, the most commonly used single-qubit gates are the rotation gates R_x -

gate, R_y -gate, and R_z -gate [23]. They rotate the qubit around the Pauli-X axis, Pauli-Y axis, and Pauli-Z axis of the Bloch sphere, respectively, measured in radians. Compared to the other gates, R_y -gate is easier to implement and manipulate. x_{ij} is the classical data to be encoded. The encoding method preserves the advantages of the hybrid amplitude encoding and no longer performs independent normalization on the data of each qubit block. Instead, it standardizes all classical data via scaling it within the range of $[0, \pi]$, thereby avoiding the problem of data representation distortion.

The tree-structured hybrid amplitude encoding divides the N qubit system into m independent blocks of n qubits, and each block can encode $2^n - 1$ bits of classical data in a tree amplitude encoding manner, and the entire N -qubit system can encode $m(2^n - 1)$ bits of classical data. Tree-structured hybrid amplitude encoding offers a high level of flexibility, which can achieve a balance between the depth and width of the quantum circuit by adjusting the parameters m and n . Additionally, uniform normalization of classical data across all amplitude encodings is employed to prevent data representation distortion resulting from inconsistent scaling. This optimization ensures more robust and accurate quantum computations.

3.3. Circuit design of quantum convolutional neural network

The overall structure of the QCNN based on variational quantum circuit is typically similar to that of the QCNN [24] designed by Lü et al.,

Table 1

Average accuracy values of the QCNN in classifying the MNIST datasets under different conditions.

Circuit	Pre-processing	Angle encoding	Dense angle encoding	Amplitude encoding	Hybrid amplitude encoding $n = 2$	Hybrid amplitude encoding $n = 4$	Our encoding $n = 2$	Our encoding $n = 4$
1	PCA	97.96%	96.01%	95.25%	66.47%	73.50%	97.87%	96.81%
	Autoencoder	92.86%	86.10%		89.70%	94.77%	91.74%	88.69%
2	PCA	98.21%	95.18%	97.31%	68.94%	74.26%	98.21%	97.68%
	Autoencoder	92.89%	90.81%		85.82%	92.23%	91.11%	88.67%
3	PCA	97.81%	98.25%	94.28%	72.91%	78.39%	98.07%	97.68%
	Autoencoder	93.62%	94.20%		92.01%	90.77%	92.08%	89.48%
4	PCA	98.37%	98.47%	92.72%	76.62%	77.21%	98.11%	98.14%
	Autoencoder	97.78%	93.69%		90.37%	91.14%	92.21%	89.45%
5	PCA	98.44%	97.71%	92.67%	69.13%	73.38%	98.11%	98.12%
	Autoencoder	92.68%	95.11%		90.18%	92.94%	91.82%	89.77%
6	PCA	98.56%	98.51%	95.91%	76.22%	74.98%	98.18%	98.08%
	Autoencoder	96.09%	98.66%		90.14%	93.97%	91.79%	88.83%
7	PCA	98.25%	98.51%	98.35%	78.30%	76.83%	98.21%	98.23%
	Autoencoder	98.61%	97.81%		93.93%	94.05%	94.44%	91.59%
8	PCA	98.23%	97.81%	97.05%	77.87%	79.24%	98.14%	98.40%
	Autoencoder	94.56%	94.49%		92.42%	93.59%	94.79%	91.05%

Table 2

Average accuracy values of the QCNN in classifying the Fashion-MNIST datasets under different conditions.

Circuit	Pre-processing	Angle encoding	Dense angle encoding	Amplitude encoding	Hybrid amplitude encoding $n = 2$	Hybrid amplitude encoding $n = 4$	Our encoding $n = 2$	Our encoding $n = 4$
1	PCA	84.63%	82.36%	90.77%	58.60%	64.51%	87.13%	82.52%
	Autoencoder	86.58%	88.25%		85.56%	80.15%	83.03%	87.28%
2	PCA	88.75%	85.40%	89.40%	59.35%	63.13%	88.13%	86.27%
	Autoencoder	91.62%	91.74%		91.64%	94.05%	93.00%	92.83%
3	PCA	88.08%	87.43%	89.57%	62.51%	64.25%	85.02%	86.53%
	Autoencoder	91.68%	91.07%		90.55%	88.41%	92.27%	91.83%
4	PCA	87.35%	88.23%	91.38%	64.55%	66.77%	89.88%	86.85%
	Autoencoder	94.1%	93.87%		91.94%	92.52%	92.25%	91.98%
5	PCA	88.47%	87.23%	91.62%	65.87%	66.19%	89.25%	86.43%
	Autoencoder	93.17%	92.42%		90.64%	91.45%	92.75%	91.65%
6	PCA	89.60%	90.19%	89.70%	63.69%	67.22%	89.55%	87.32%
	Autoencoder	93.50%	93.75%		92.54%	90.26%	90.88%	91.80%
7	PCA	90.43%	91.85%	89.90%	65.15%	68.74%	88.07%	90.00%
	Autoencoder	92.71%	94.18%		91.19%	91.35%	93.33%	91.68%
8	PCA	90.15%	89.00%	90.68%	66.34%	67.36%	89.43%	89.75%
	Autoencoder	94.05%	93.30%		92.68%	92.46%	93.75%	92.75%

including data encoding, convolution pooling and full connection layer. However, for our specific experiment, which involves binary classification, the fully connected layer is not necessary. Therefore, the QCNN model designed does not include the fully connected layer. The QCNN circuit structure of 8 qubits used in experiment is shown in Fig. 3.

The quantum circuits for both convolution and pooling operations play a crucial role in determining the classification performance of the QCNN. Fig. 4 illustrates different convolutional kernel circuits used for training the QCNN, each with a varying number of parameters. Circuit 1 is called tree tensor network [25], while circuit 7 is a special unitary group SU(4) [26].

The pooling layer applies parameterized quantum gates on two qubits and continues to use only one of them, achieving the goal of reducing the qubits by half. The pooling layer is shown in Fig. 5.

4. Simulation results and performance analyses

In the simulation experiment, we will compare the effects of PCA and autoencoder as dimensionality reduction methods. These two techniques will be evaluated on various datasets, and their performances will

Table 3

Comparison of accuracy with the QCNN in classifying the MNIST Datasets.

QCNN model	Qubit number	Number of parameters	Maximal accuracy
[24]	6	54	96.65%
QCNN (Circuit 1)	8	12	97.92%
QCNN (Circuit 8)	8	51	98.72%

be assessed with different structural variational quantum circuits and datasets. Since the NISQ device only supports a small number of qubits, the variational quantum circuit designed in this experiment is fixed at 8 qubits.

The convolutional neural network structure itself has a dimensionality reduction process. With three convolutional layers and pooling layers, the information carried by the 8-qubit circuit is extracted into one qubit. The loss function selected for the experiment is the cross entropy function, and the binary cross entropy function can be expressed as

$$L_C(\theta) = -\frac{1}{2m} \sum_{i=1}^m [y_i \ln C(x_i; \theta) + (1 - y_i) \ln(1 - C(x_i; \theta))] \quad (4)$$

Table 4

Comparison of classification accuracy between QCNN and CNN.

Model	Dataset	PCA-8	Autoencoder-8	PCA-16	Autoencoder-16
CNN	MNIST	86.32%	94.28%	96.04%	91.76%
	Fashion-MNIST	87.67%	90.43%	85.37%	93.38%
QCNN (Circuit 5)	MNIST	97.61%	94.61%	98.20%	95.78%
	Fashion-MNIST	88.86%	93.12%	87.23%	93.66%
QCNN (Circuit 6)	MNIST	98.54%	96.35%	98.64%	95.88%
	Fashion-MNIST	89.65%	92.94%	88.91%	93.09%

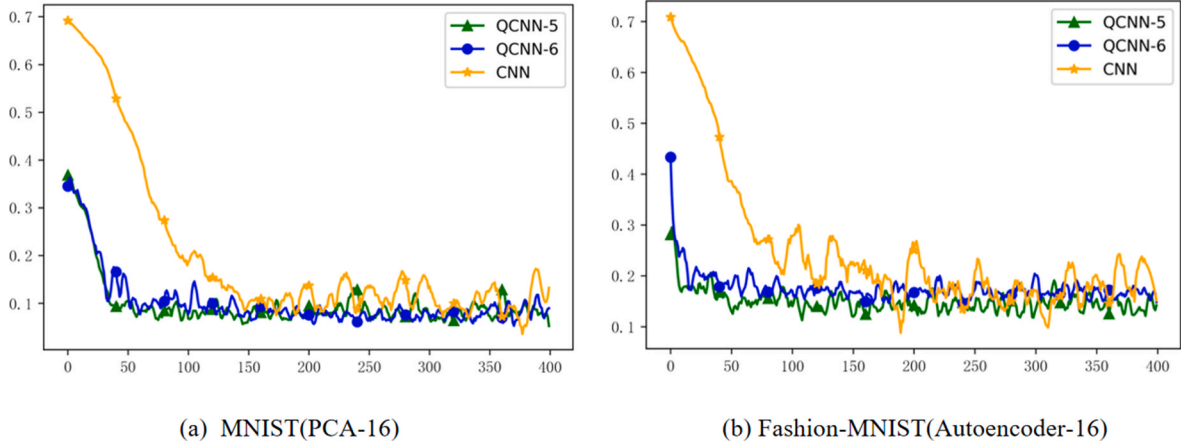


Fig. 6. Convergence of loss function with QCNN and CNN in the training process of different dataset classification tasks.

where C represents the QCNN, m represents batch size, and $y_i \in \{0, 1\}$ is the label corresponding to x_i . The loss function $L_C(\theta)$ is minimized by the Nesterov momentum optimizer provided by PennyLane framework. The learning rate is set to a default value of 0.01, and the number of iterations is set as 200. The number of hidden layers of the autoencoder is set to 1 to guarantee the data pre-processing process is not too complex.

4.1. Classification experiment under MNIST dataset

The classification experiments on the MNIST handwritten digit image dataset are conducted three times under various conditions. The average accuracy results for these experiments are shown in Table 1. The serial numbers of the quantum convolutional layer circuits correspond one-to-one with those depicted in Fig. 4. For the amplitude encoding experiment, it's important to note that amplitude encoding allows the largest amount of data to be encoded with the same width of quantum circuit. In the case of an 8-qubit circuit, it can encode 256 bits of classical data. Therefore, in the amplitude encoding experiment, the image data was not processed with dimensionality reduction, but the image size was directly reduced to 16×16 , and then encoded into the quantum circuit. In the angle encoding experiment, both dimensionality reduction methods reduced the data dimension to 8. Additionally, in the dense angle encoding experiment, the data dimension was reduced to 16. It's worth mentioning that in an 8-qubit circuit, both hybrid amplitude encoding and hybrid tree amplitude encoding have two variants. One is to divide the entire circuit into four 2-qubit blocks, labeled by $n = 2$, the other is to divide it into two 4-qubit blocks, labeled by $n = 4$.

Regarding data dimension reduction with PCA, both angle coding and dense angle coding demonstrate excellent and consistent performance. These methods do not impose stringent requirements on the choice of the quantum circuit. However, when utilizing autoencoder for data dimensionality reduction, the two angle encoding techniques exhibit relatively lower stability, resulting in lower average accuracy compared to PCA. The highest accuracy achieved in all experiments is 98.66%, and it is obtained by combining Circuit 6, autoencoder-based dimension reduction, and dense angle coding. The reason for the unstable performance of the autoencoder may be its simple structure and insufficient training iterations. Therefore, for simpler tasks, it is more appropriate to choose the PCA dimensionality reduction method, which offers stable performance and consumes fewer resources.

From the perspective of convolutional layer architecture, the parameters in the convolutional circuits gradually increase from Circuit 1 to Circuit 8. An increase in parameters signifies enhanced quantum entanglement and learning capabilities. Notably, as the number of parameters reaches its maximum, the overall accuracy consistently improves. For instance, when combining Circuit 1 with PCA-based

dimension reduction and our coding approach under $n = 4$, the accuracy is 96.81%. In similar settings, Circuit 2 and Circuit 3, each with 4 parameters, achieve accuracies of 97.68% and 97.68%, respectively. Circuits 4 and 5, equipped with 6 parameters, attain accuracies of 98.14% and 98.12%, respectively. While Circuit 6, featuring 10 parameters, achieves an accuracy of 98.08%. Finally, Circuits 7 and 8, both employing 15 parameters, reach impressive accuracies of 98.23% and 98.40%, respectively.

4.2. Classification experiment under Fashion-MNIST dataset

In quantum machine learning for image processing, it's common to use either basic MNIST datasets or even simpler self-created datasets to assess classification performance. However, overly simplistic datasets may not effectively gauge algorithm performance in practical applications. Therefore, to conduct more comprehensive verification experiments, we employed the Fashion MNIST dataset, as shown in Table 2. The numbers of the convolutional layer circuits also correspond one-to-one to those in Fig. 4. In the amplitude encoding experiment, the data preprocessing method still directly reduces the image size.

Table 2 indicates a slight decrease in overall accuracy when using the Fashion-MNIST dataset instead of the MNIST dataset. This drop in accuracy can be attributed to the higher complexity of the Fashion-MNIST images, which inherently increases the difficulty of the classification task.

In experiments involving angle encoding and dense angle encoding, accuracy tends to increase as the number of parameters grows. However, in hybrid amplitude encoding experiments, the data dimensionality reduction performance of PCA is significantly inferior, even if dropping to below 60%. This is primarily because PCA struggles to handle complex image data, and hybrid amplitude encoding separately normalizes the encoded classical data for each qubit region, resulting in inconsistent scaling ratios across different regions. This inconsistency hinders the accurate representation of the original data. Conversely, autoencoders demonstrate relatively better performance when dealing with complex image data and dimensionality reduction. Hybrid tree amplitude encoding consistently exhibits stable encoding performance across various conditions, closely approaching the accuracy achieved with angle encoding. This is because it applies uniform normalization to all qubit regions, allowing for a relatively accurate representation of the original data.

Comparing the classification accuracy of the QCNN on the MNIST and the Fashion-MNIST datasets reveals several insights. PCA exhibits stable dimensionality reduction performance and consumes fewer computational resources. However, PCA is limited to simple linear calculations and may not effectively handle complex data. On the other

hand, even if with simplified design and training to save computational resources in the experiment, autoencoders still perform well when processing complex data. Therefore, autoencoders are better suited for complex image classification tasks. Angle encoding, dense angle encoding, amplitude encoding, and hybrid tree amplitude encoding have all demonstrated relatively stable performance in different experiments. The choice of encoding method can be made based on specific situations and the requirements of the task at hand.

4.3. Comparison with similar works

The proposed QCNN structure is similar to the pure variational component sub-circuit QCNN designed by Lü et al. [24] for image classification. In our classification experiments on the MNIST dataset, as shown in Table 3, Lü et al. reduced the original data dimension to 64 using PCA and encoded the data into a quantum circuit via amplitude encoding. In contrast, our QCNN, based on Circuit 1 and Circuit 8 in Table 3, reduced the original data dimension to 30 using PCA and employed a hybrid tree amplitude encoding method to encode the data into the quantum circuit. This adaptation allows for a more tailored and potentially improved approach to handling data dimensionality reduction and encoding in quantum image classification tasks. It can be seen from Table 3 that the designed QCNN can provide higher classification accuracy with fewer parameters than the QCNN in Ref. [24]. One significant factor contributing to the differences in results is the variation in quantum state preparation methods. In an effort to conserve resources during amplitude encoding, Lü et al. employed an approach that reduced the time needed for data encoding and the depth of the quantum circuit by approximating the coding data with a trainable model. However, this approach still resulted in inaccuracies in data representation. To strike a balance between reducing the depth of the quantum circuit and maintaining the accuracy of data representation, we opted for the hybrid tree amplitude encoding method. This encoding method allows us to encode the data in a manner that achieved higher classification accuracy while mitigating the limitations associated with data representation.

Moreover, to perform a scientifically controlled comparison between QCNN and CNN, it is crucial to ensure that the number of parameters in both models is either close or comparable. When dealing with high-dimensional input data, training a CNN with a small number of parameters becomes challenging. Therefore, dimensionality reduction processing is necessary for the input data of the CNN to bring it to a manageable size for a fair and meaningful comparison with QCNN. The data dimension is reduced to 8 and 16 with PCA and Autoencoder, respectively. If the dimension of input data of the model is 8, the number of the parameters involved in the CNN model is 26, while if the dimension of input data is 16, the number of parameters in the CNN model is 34. The QCNN of the control group selected Circuit 5 and Circuit 6 with the closest number of parameters 24 and 36, respectively. The results of the scientific control are shown in Table 4. The learning rate and the number of iterations in the experiment is set as 0.01 and 400, respectively.

The experimental results are the average results of five rounds of experiments, where PCA- n and Autoencoder- n represent the data dimension reduction method used and the data dimension size obtained after dimension reduction.

Under similar experimental conditions and with a comparable number of parameters, QCNN consistently outperforms CNN in terms of classification accuracy. This superiority can be attributed to the convolutional layer in QCNN, which entangles all qubits and possesses the potential to extract global features from the data. In contrast, classical convolutional layers in CNNs are limited to extract local features. This distinction gives QCNN an advantage in capturing and utilizing more extensive and holistic information for classification tasks.

The convergence of loss function in some experimental training processes is shown in Fig. 6, where QCNN-5 and QCNN-6 represent the

QCNN with Circuit 5 and Circuit 6 respectively. For the convenience of observation, the loss function curve has been smoothed to some extent. In the MNIST classification experiment, PCA-16 group with good CNN performance was selected, while in the Fashion MNIST classification experiment, Autoencoder-16 group with good CNN performance was selected. It's evident from the convergence of the loss function that QCNN exhibits a faster rate of convergence and greater training stability across various tasks. This advantage can be attributed to the global entanglement property of the quantum circuit. The experimental results demonstrate that QCNN not only delivers more accurate classification results but also achieves a quicker rate of convergence during training.

5. Conclusion

A quantum machine learning algorithm for image classification is investigated based on the most popular convolutional neural network in deep learning and quantum computing. A universal QCNN model structure is designed based on the structure of classical CNN models, and its classification performances in different situations are evaluated. Combining the advantages of angle encoding and amplitude encoding, a more flexible hybrid tree-structured amplitude encoding method is proposed. This encoding method can flexibly adjust the width and the depth required for the data encoding circuit according to different real tasks and quantum device limitations. The experimental results indicate that the hybrid tree-structured amplitude encoding performs well and stably under different situations. In terms of learning speed and classification accuracy, QCNN outperforms CNN significantly. Compared to CNN, QCNN achieves the accuracy rates of 98.40% and 93.75% on the MNIST dataset and the Fashion-MNIST dataset classification tasks respectively, whereas CNN achieves the accuracy rates of 96.04% and 93.38% on the same datasets. As data dimensions continue to increase, quantum datasets may replace traditional datasets, making the development of quantum classifiers increasingly important. We also look forward to further improving and optimizing our QCNN model to achieve better performance on more complex tasks and datasets.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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